

A MICROSCOPY-BASED APPROACH TO CHARACTERIZING PARTICULATE CONTAMINANTS IN SNOWFALL ON SUBURBAN LONG ISLAND, NEW YORK



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Abstract:

Snowfall can act as an effective sampler of atmospheric particulates, capturing a wide range of contaminants that are later released into the environment during snowmelt. This study provides a preliminary microscopic characterization of particulate contaminants in suburban snowfall, establishing a methodological framework for future geochemical analysis of heavy metal deposition in winter precipitation.

Snow samples were collected from a suburban site on Long Island, specifically the rooftop of Gittleston Hall at Hofstra University in Hempstead, New York, during two snowfall events in 2026. Controlled collection procedures were used to minimize contamination, followed by melting, filtration, and laboratory preparation.

Collected particulates were analyzed using optical microscopy and scanning electron microscopy (SEM) to examine morphology, size distribution, and aggregation patterns. Observations revealed a heterogeneous mixture of contaminants, including fine particles, fibrous materials, and irregularly shaped aggregates. SEM imaging identified clustered and porous structures, suggesting the presence of complex, multi-source atmospheric particulates likely influenced by urban and suburban activities.

Although elemental composition was not determined in this phase of the study, the observed diversity of particles indicates multiple potential sources, including anthropogenic emissions (such as road salt) and natural debris. These findings support the role of snowfall as a temporary reservoir for atmospheric contaminants and highlight its importance in transporting particulate matter into terrestrial and aquatic systems during melt events.

Background Information:

- Snowfall recognized as efficient collector of atmospheric particulate matter
- Worldwide studies demonstrating snow can accumulate variety of pollutants, including particulate matter, black carbon, road salts, and heavy metals
- Previous research in Europe, Asia, and North America has shown that snowpack composition reflects local emission sources
- Limited site-specific characterization of particulate contaminants in snowfall across suburban environments such as Long Island, New York
- Study presents a preliminary investigation of particulate contaminants collected from snowfall at a suburban site in Hempstead, New York (Hofstra University campus)

Sample Collection



Figure 1: Snow sampling procedure on the rooftop of Gittleston Hall, Hofstra University (Hempstead, NY) during the January 28, 2026 snowfall event. Samples were collected from the under the snowpack to minimize contamination from surface runoff and post-depositional atmospheric inputs.

- Sufficient particulate material collected for analysis
- Minor preparation issues did not affect sample quality

Method/Materials:

Study Population & Focus

- Snowfall collected from the rooftop of Gittleston Hall at Hofstra University
- Samples taken during two separate snowfall events in 2026

Sample Collection

- Snow gathered from the middle layer of the snowpack to avoid surface contamination
- Clean tools used to keep samples consistent and reduce outside debris

Laboratory Preparation

- Snow was melted, filtered, and concentrated for microscopy work
- A simple filtration setup used to collect particulate material for analysis

Microscopy & Analysis

- Optical microscopy used to document particle size, shape, and texture
- SEM imaging used to examine clustered and irregular particles in more detail



Figure 2: Schematic diagram of the filtration setup used for preliminary sample processing, illustrating the modified container and filter system used to collect and concentrate particulate matter from melted snow samples.

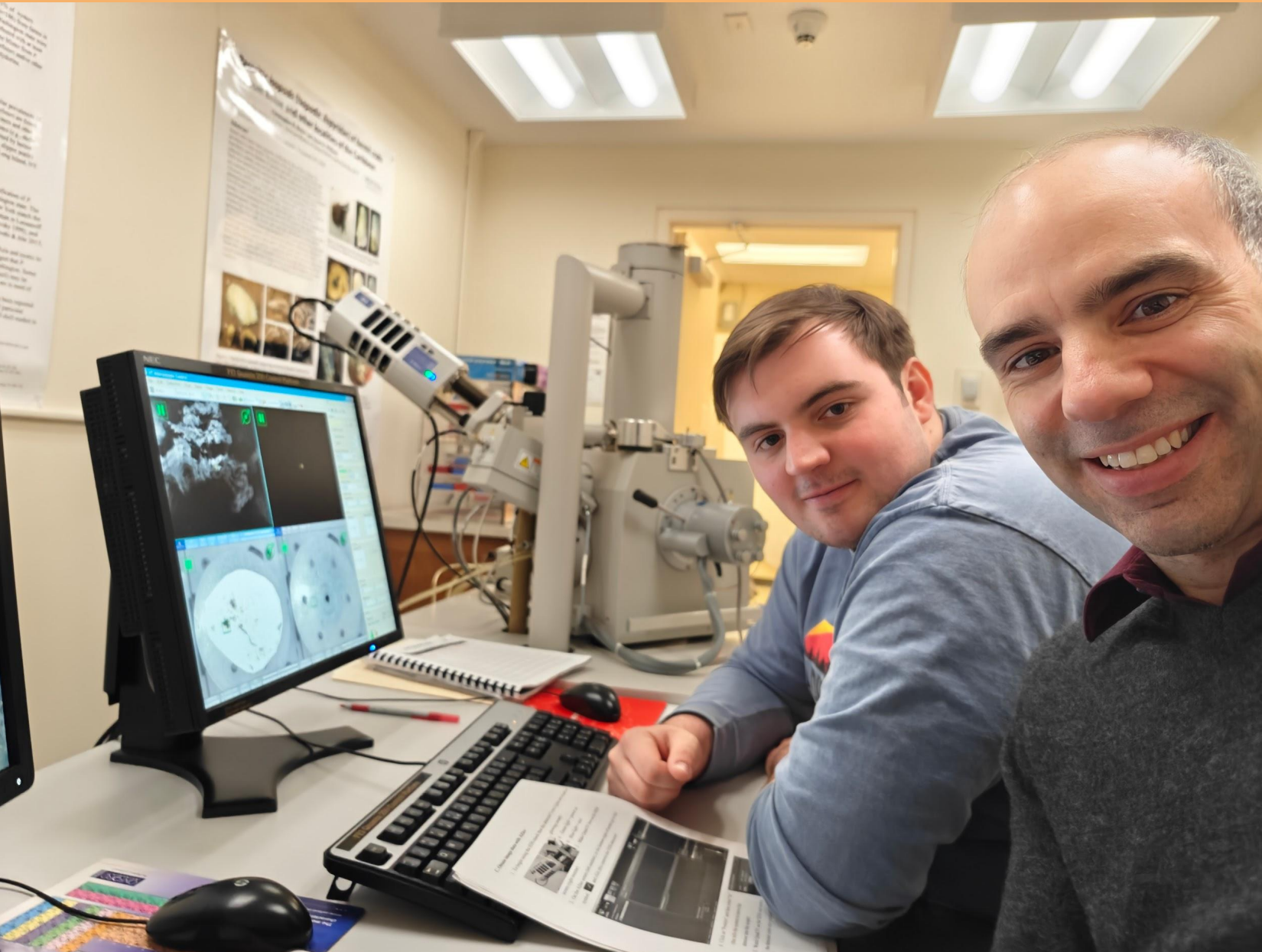


Figure 3: Dominick Colucci (left) and Dr. Antonios Marsellos conducting preliminary analysis of snow-derived particulate contaminants using a scanning electron microscope (SEM) at Hofstra University.

Results:

Optical Microscopy

- Heterogeneous particles observed:
 - Dark aggregates (dominant)
 - Fibrous material
 - Granular debris
- Particle size range: <100 μm to >500 μm
- Likely sources:
 - Organic/textile material (dark aggregates)
 - Mineral dust (granular particles)
 - Road-derived materials (salt, asphalt) $\rightarrow$ anthropogenic input

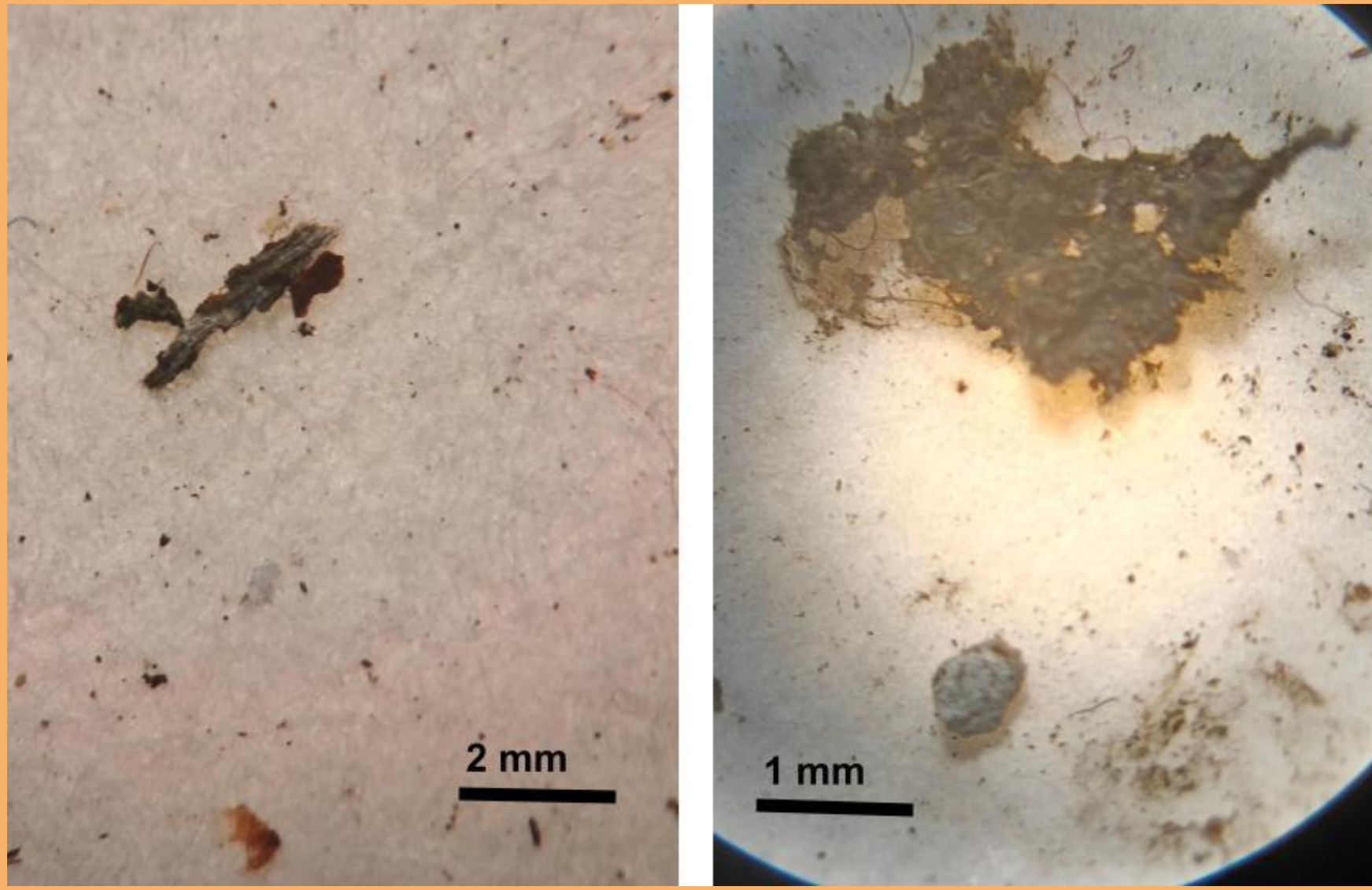


Figure 4: Optical microscopy images of particulate contaminants extracted from snowfall samples, showing heterogeneous material including dark aggregated particles, fibrous debris, and granular components. Scale bars indicate millimeter-scale variation in particle size.

SEM Analysis

- Particles characterized by:
 - Irregular, non-spherical shapes
 - Porous, clustered structures
 - Filamentous textures
- No well-defined crystalline structures
- Indicates:
 - Mixed particle types
 - Secondary aggregation processes
 - Urban atmospheric contamination

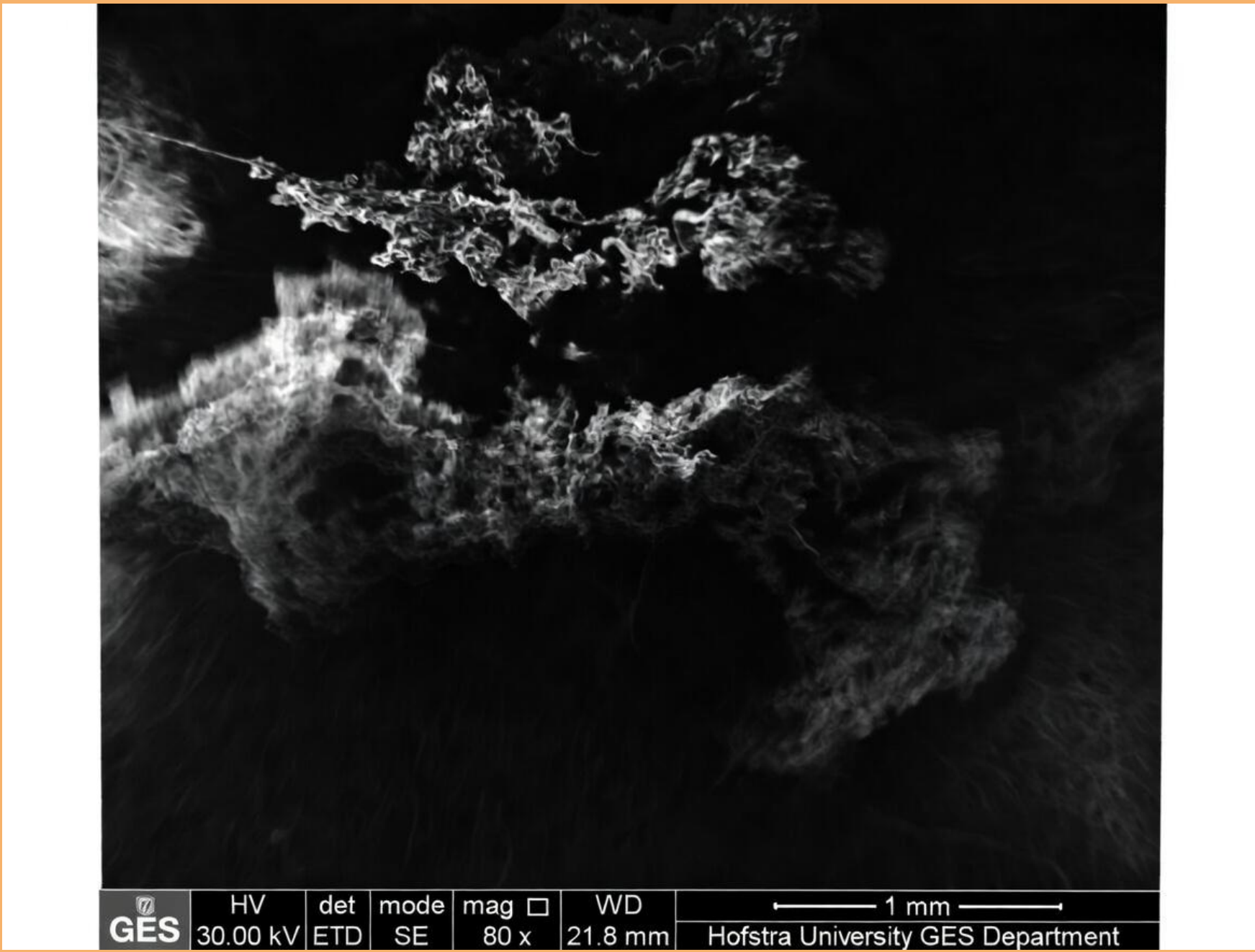


Figure 5: SEM Result of the same particle of figure 4 (right). SEM image shows aggregated particulate matter from snowfall samples, revealing irregular, non-spherical morphology with porous, clustered structures and filamentous textures.

Discussion:

Contamination Sources

- Evidence of mixed origins:
 - Atmospheric deposition
 - Urban dust
 - Organic debris
 - Anthropogenic inputs
- Snowfall acts as an efficient atmospheric sampling, capturing both anthropogenic and natural particulates
- The variety of particles showed that the snow collected contained contaminants from both human and natural sources
- Snowfall acted as a temporary reservoir

Conclusion:

- Snowfall acted as a short-term reservoir for particulates
 - Collected particles from the surrounding suburban environment
- Snowfall captured particles from multiple sources
 - Manmade sources
 - Natural environmental sources
- The study created a useful analysis framework
 - showed that optical microscopy and SEM can identify snowfall contaminants

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Credit Authorship Contribution Statement:

Colucci, D: snow sample collection, set up laboratory procedures, performed laboratory procedures, microscope procedure, SEM scanning, writing-Abstract, Introduction, Methods, Results, Discussion, Conclusion, editing
Schwartz, J: set up laboratory procedures, performed laboratory procedures, writing- Abstract, Introduction, Methods, Results, Discussion, Conclusion, editing
Singh, I: writing-Abstract, Introduction, Methods, Results, Discussion, Conclusion, editing
Marsellos, A.E.: supervision, research mentorship, sampling guidance.

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